Lec-3

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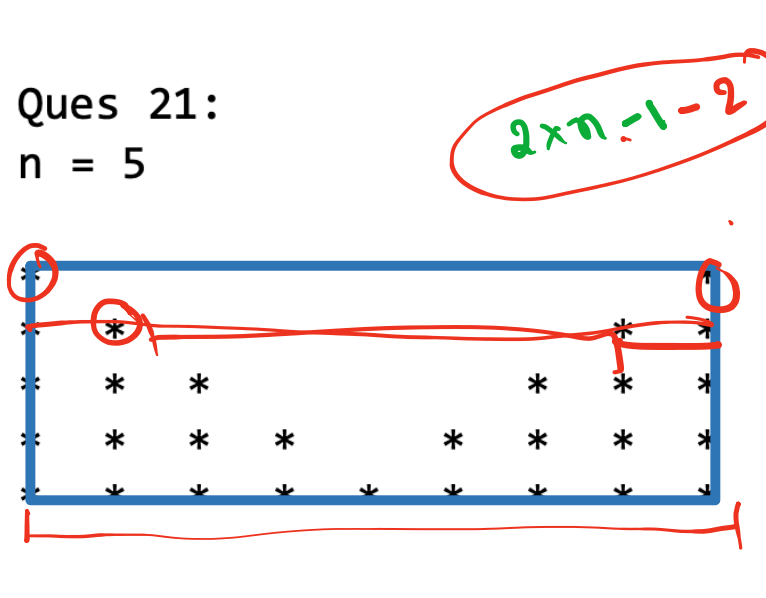


Ques 21:
n = 5

```
*                       *
*   *               *   *
*   *   *       *   *   *
```

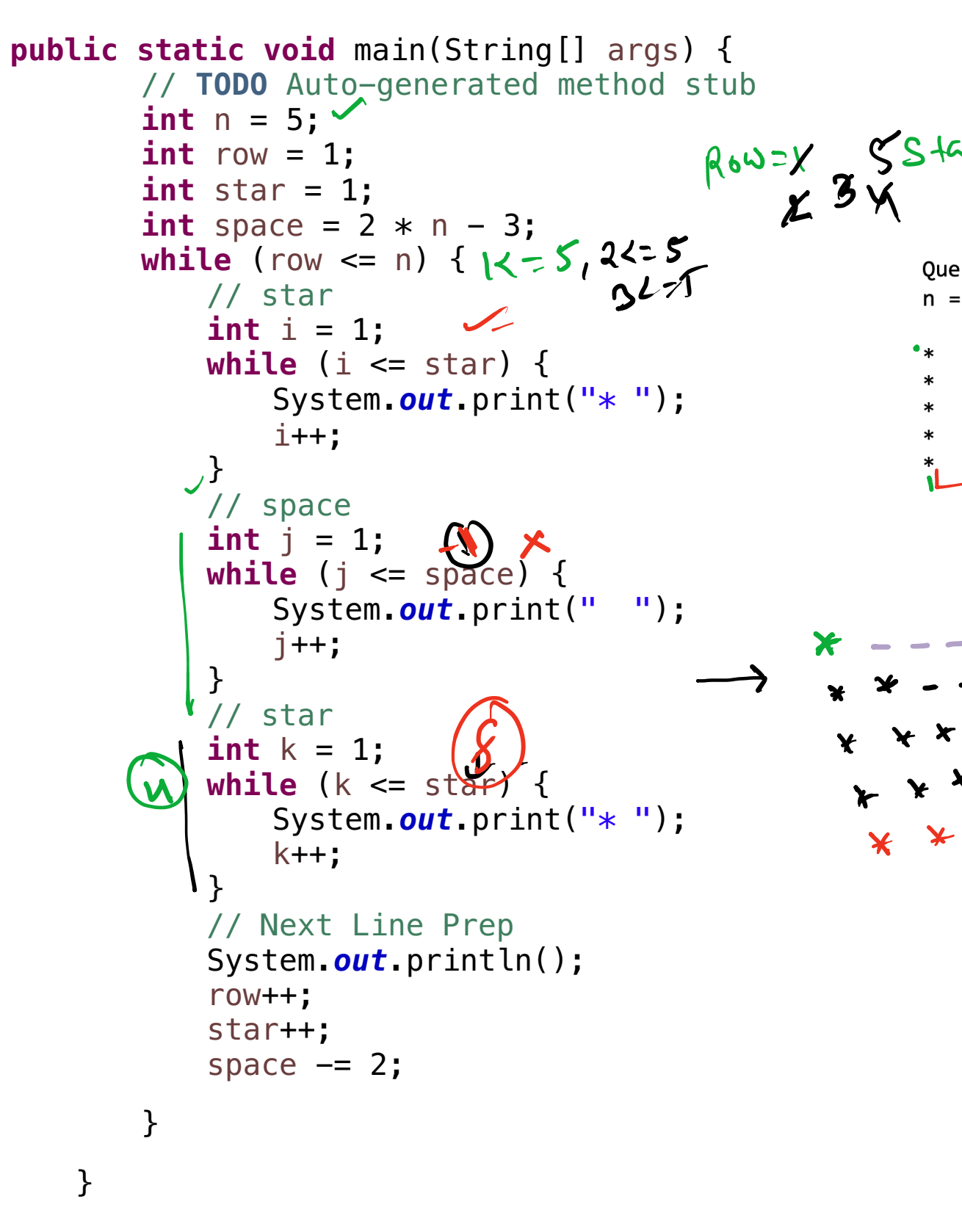
2×n-1-2

2×n-3

Row=1 star=1 space=2×n-3

while (Row ≤ n) {
// star k<=p
// space j<-
// star k<=n

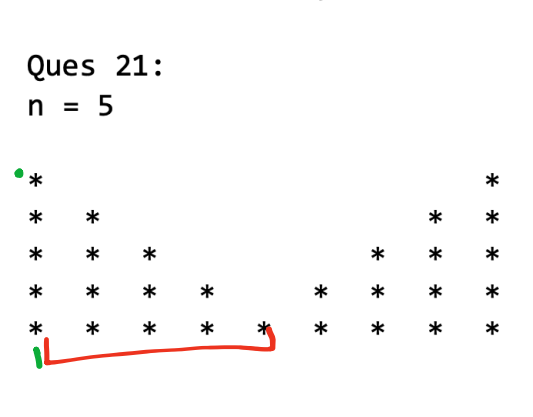
// next line ki Prep
S.o.p() ~~X~~
Row++
Star++
Spae -=2
}



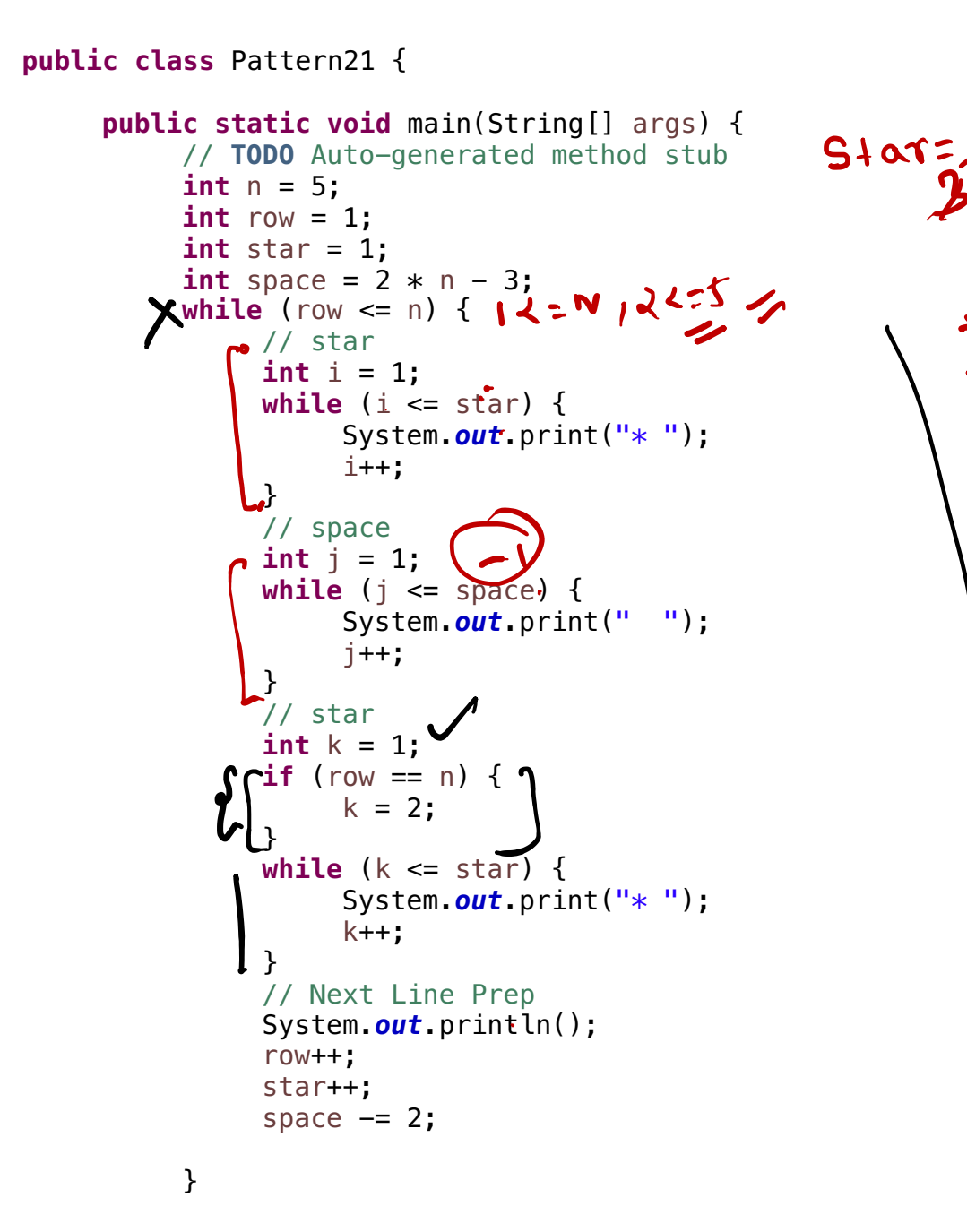
```
public static void main(String[] args) {
        // TODO Auto-generated method stub
        int n = 5;
        int row = 1;
        int star = 1;
        int space = 2 * n - 3;
        while (row <= n) {
            // star
            int i = 1;
            while (i <= star) {
                System.out.print("* ");
                i++;
            }
            // space
            int j = 1;
            while (j <= space) {
                System.out.print("  ");
                j++;
            }
            // star
            int k = 1;
            while (k <= star) {
                System.out.print("* ");
                k++;
            }
            // Next Line Prep
            System.out.println();
            row++;
            star++;
            space -= 2;

        }
    }
```

1<=5, 2<=5, 3<=5

Row=~~1~~ 2 3 4 5 Star=~~1~~ 2 3 4 5 Space=~~7~~ 5 3 1 -1



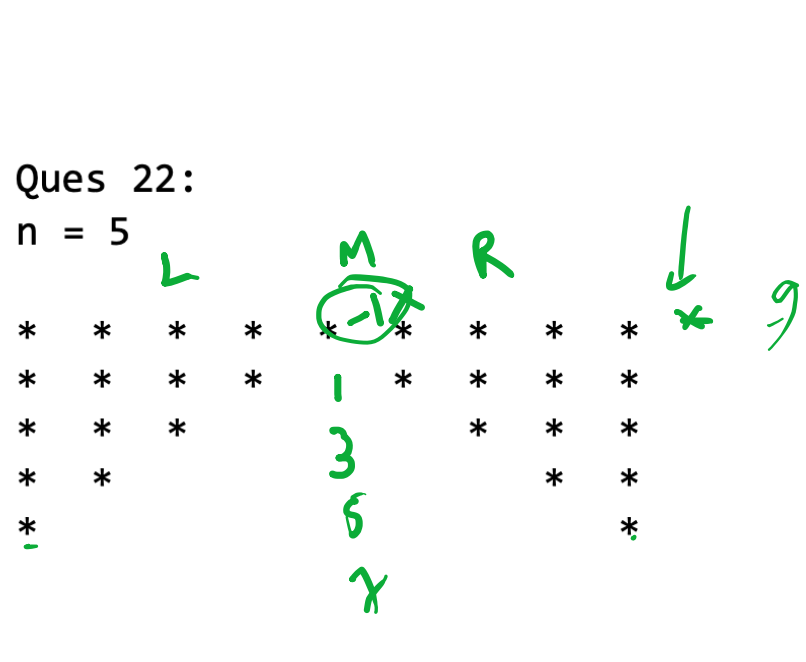
Ques 21:
n = 5



```
public class Pattern21 {

    public static void main(String[] args) {
        // TODO Auto-generated method stub
        int n = 5;
        int row = 1;
        int star = 1;
        int space = 2 * n - 3;
        while (row <= n) {
            // star
            int i = 1;
            while (i <= star) {
                System.out.print("* ");
                i++;
            }
            // space
            int j = 1;
            while (j <= space) {
                System.out.print("  ");
                j++;
            }
            // star
            int k = 1;
            if (row == n) {
                k = 2;
            }
            while (k <= star) {
                System.out.print("* ");
                k++;
            }
            // Next Line Prep
            System.out.println();
            row++;
            star++;
            space -= 2;

        }

    }

}
```

1<=N, 2<=5

Star=~~1~~ ~~2~~ 3 4 5 6 Space=~~7~~ 5 3 1 -1 Row=~~1~~ 2 3 4 5 6



Ques 22:
n = 5

```
*   *   *   *   *   *   *   *   *
*   *   *   *       *   *   *   *
*   *   *               *   *   *
*   *                       *   *
*                               *
```

L M R

9

1 3 5 7

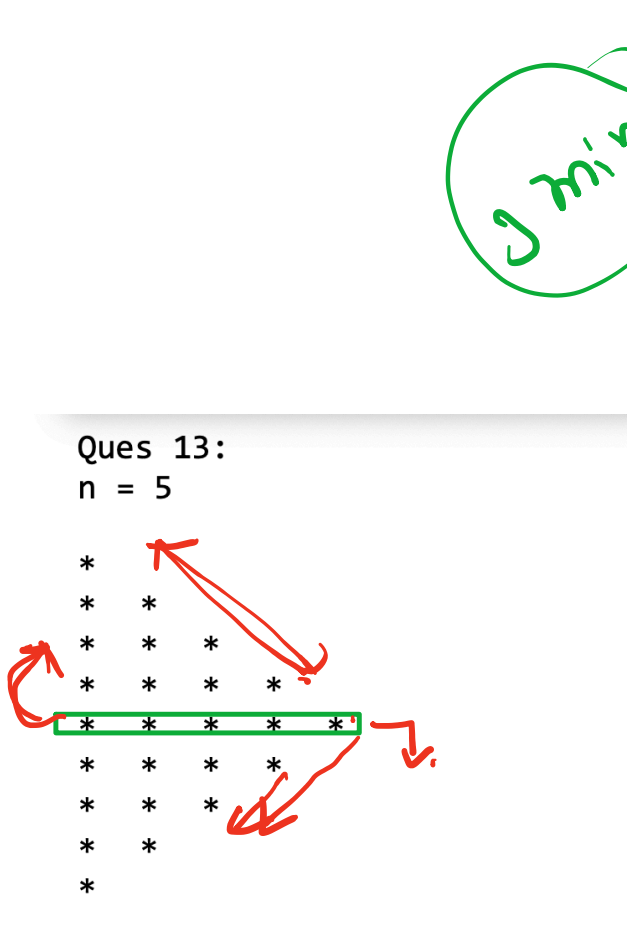
3 min

Row=1 Star=N Space=-1

while (Row ≤ n) {
// Star
// Space
// Star

next Line Prep

S.o.p()
Row++
Star--
Space+=2
}

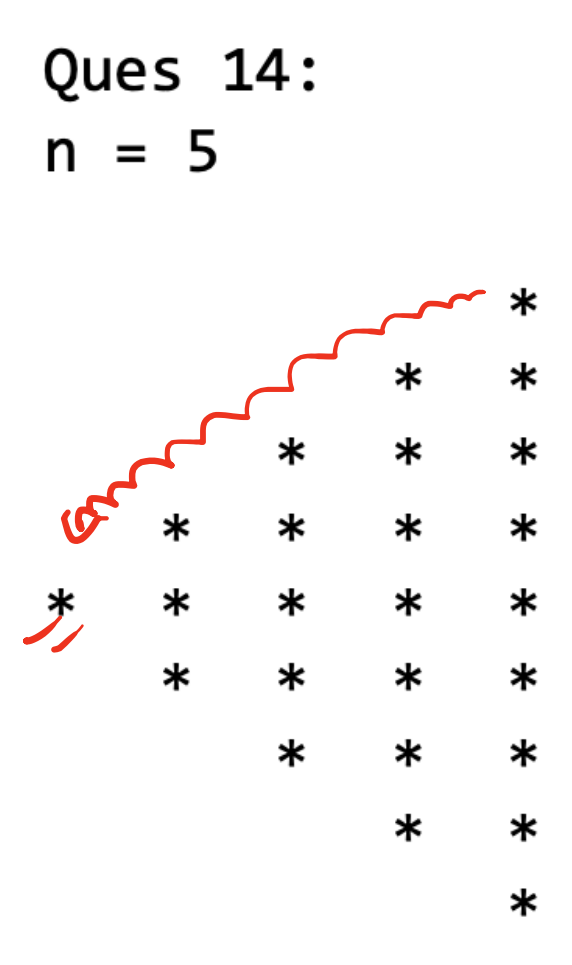


Ques 13:
n = 5

Row<=n star

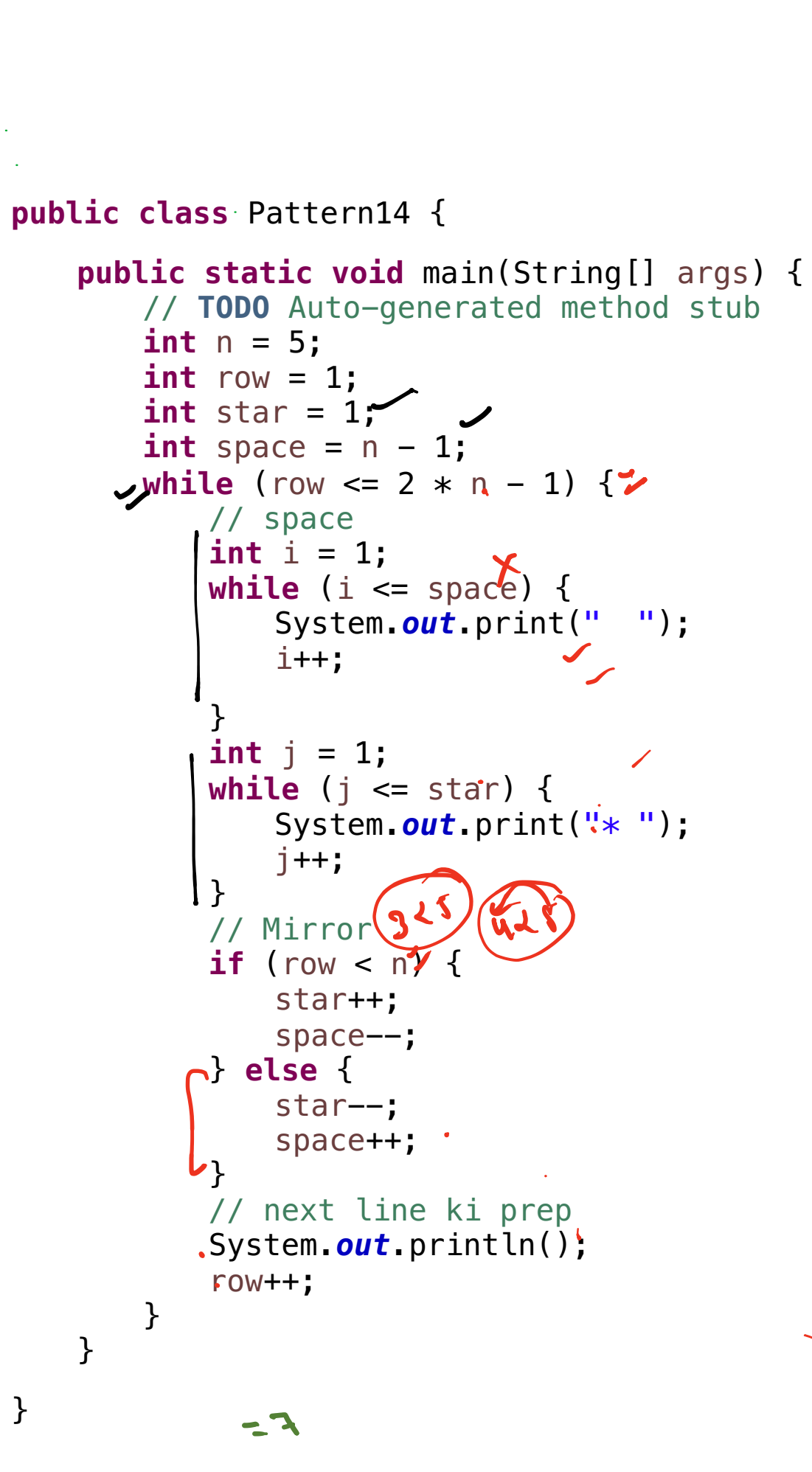
Row=1 2 ... 8 9 6 7 8 9
star= 1 2 3 4 5 4 3 2 1

while Row ≤ (2×n-1) {
// Star
int i=1
while (i<=Star) {
S.o.p ("* ")
i++
}
// mirror
if (Row < n) { → mirror peble
Star++
} else { // → mirror line या उसके बाद
Star--, =
}
// next line prep
S.o.p()
Row++
}



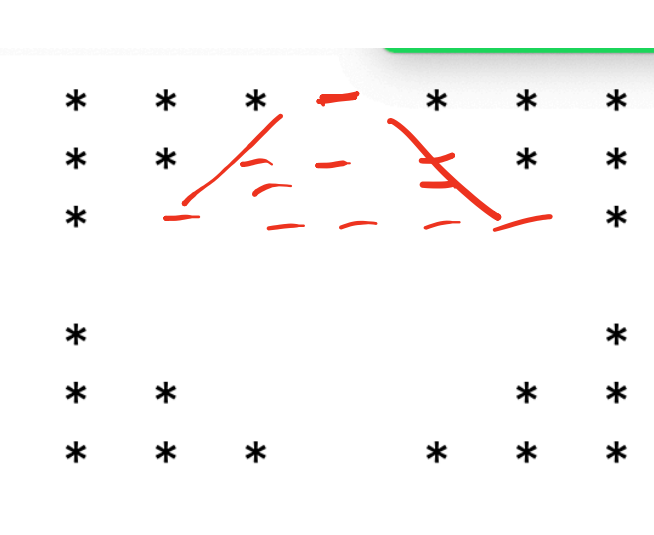
Ques 14:
n = 5

Row=1 Space=n-1 Star=1
while (Row ≤ (2×n-1)) {
// space
// star
if (Row < n) {
space--
star++
} else {
space++
star--
}
S.o.p()
Row++
}



```
public class Pattern14 {

    public static void main(String[] args) {
        // TODO Auto-generated method stub
        int n = 5;
        int row = 1;
        int star = 1;
        int space = n - 1;
        while (row <= 2 * n - 1) {
            // space
            int i = 1;
            while (i <= space) {
                System.out.print("  ");
                i++;
            }

            int j = 1;
            while (j <= star) {
                System.out.print("* ");
                j++;
            }
            // Mirror
            if (row < n) {
                star++;
                space--;
            } else {
                star--;
                space++;
            }
            // next line ki prep
            System.out.println();
            row++;
        }
    }
}
```

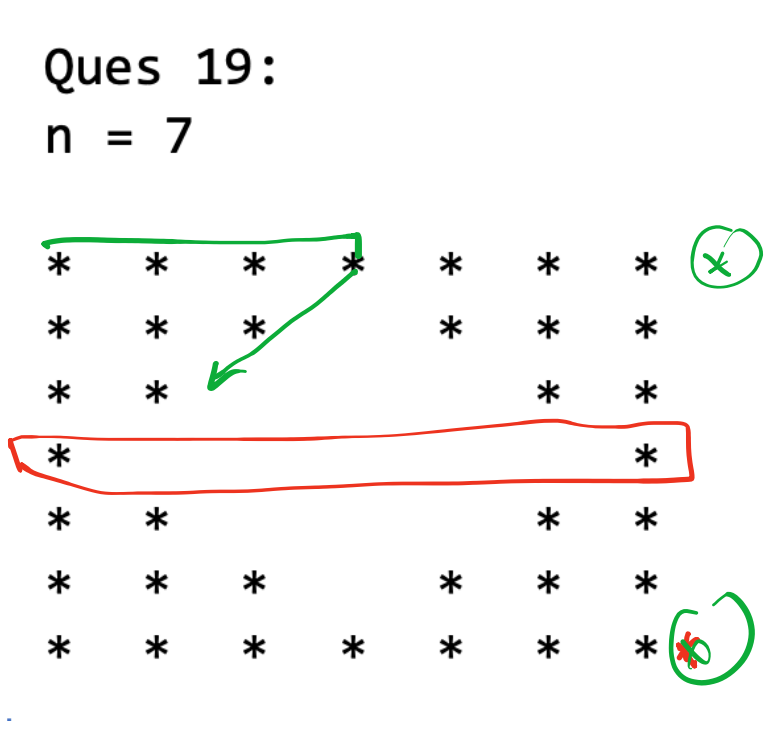
Row=~~1~~ 2 3 Star=~~1~~ ~~2~~ 3 4 5 Spce=~~4~~ 3 2 1 0 1 2



```
*   *   *       *   *   *
*   *               *   *
*                       *

*                       *
*   *               *   *
*   *   *       *   *   *
```

Row=1 Star=N/2 Space=1

while (Row ≤ n) {
// star
// space
// star
if (Row ≤ N/2 +1) {
Star--
Space+=2
} else {
Star++
Spce-=2
}
S.o.p() Row++
}

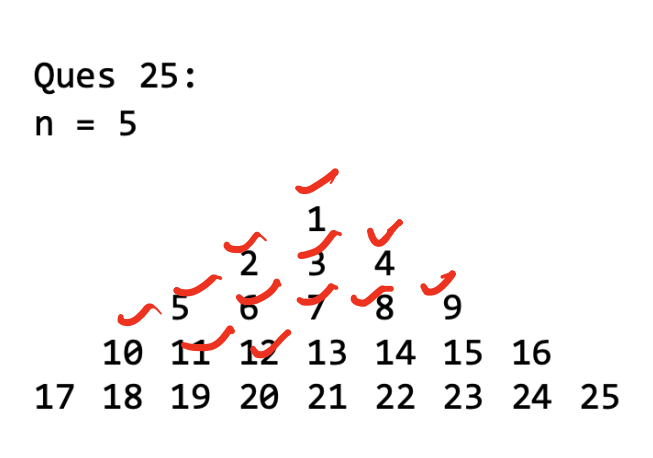


Ques 19:
n = 7

```
*   *   *   *   *   *   *
*   *   *       *   *   *
*   *               *   *
*                       *
*   *               *   *
*   *   *       *   *   *
*   *   *   *   *   *   *
```

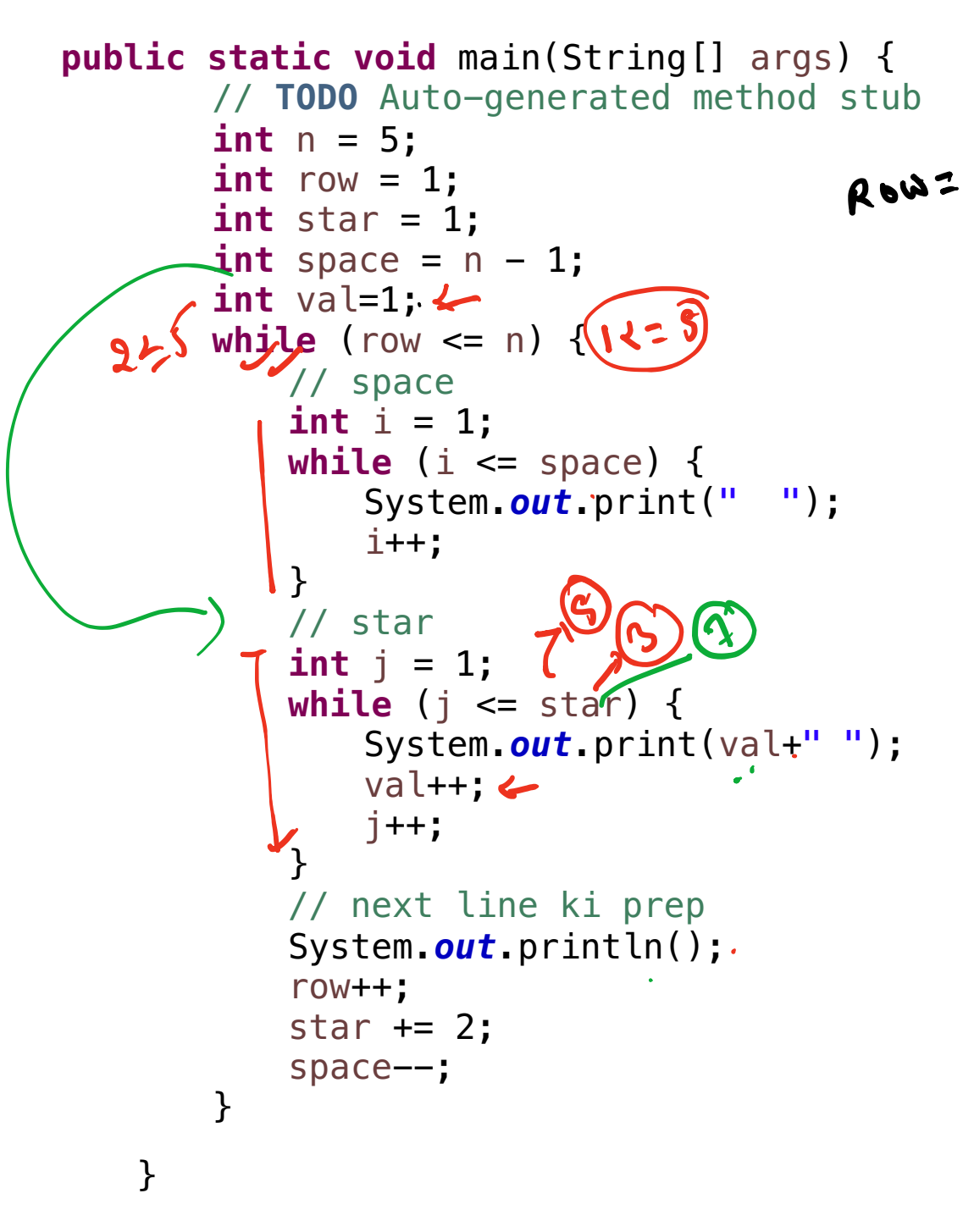
Row=1 Star=N/2 Space=-1

while (Row ≤ n) {
// star
// space
// star
if (Row < n/2 +1) {
Star--
Space+=2
} else {
Star++
Spce-=2
}
}



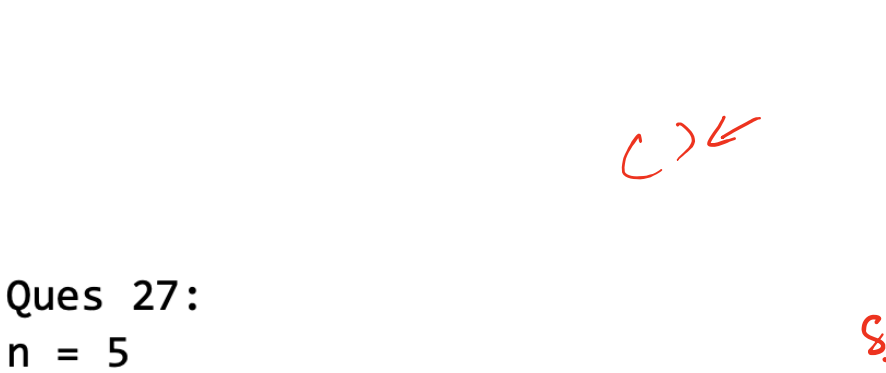
Ques 25:
n = 5

```
            1
         2  3  4
      5  6  7  8  9
   10 11 12 13 14 15 16
17 18 19 20 21 22 23 24 25
```



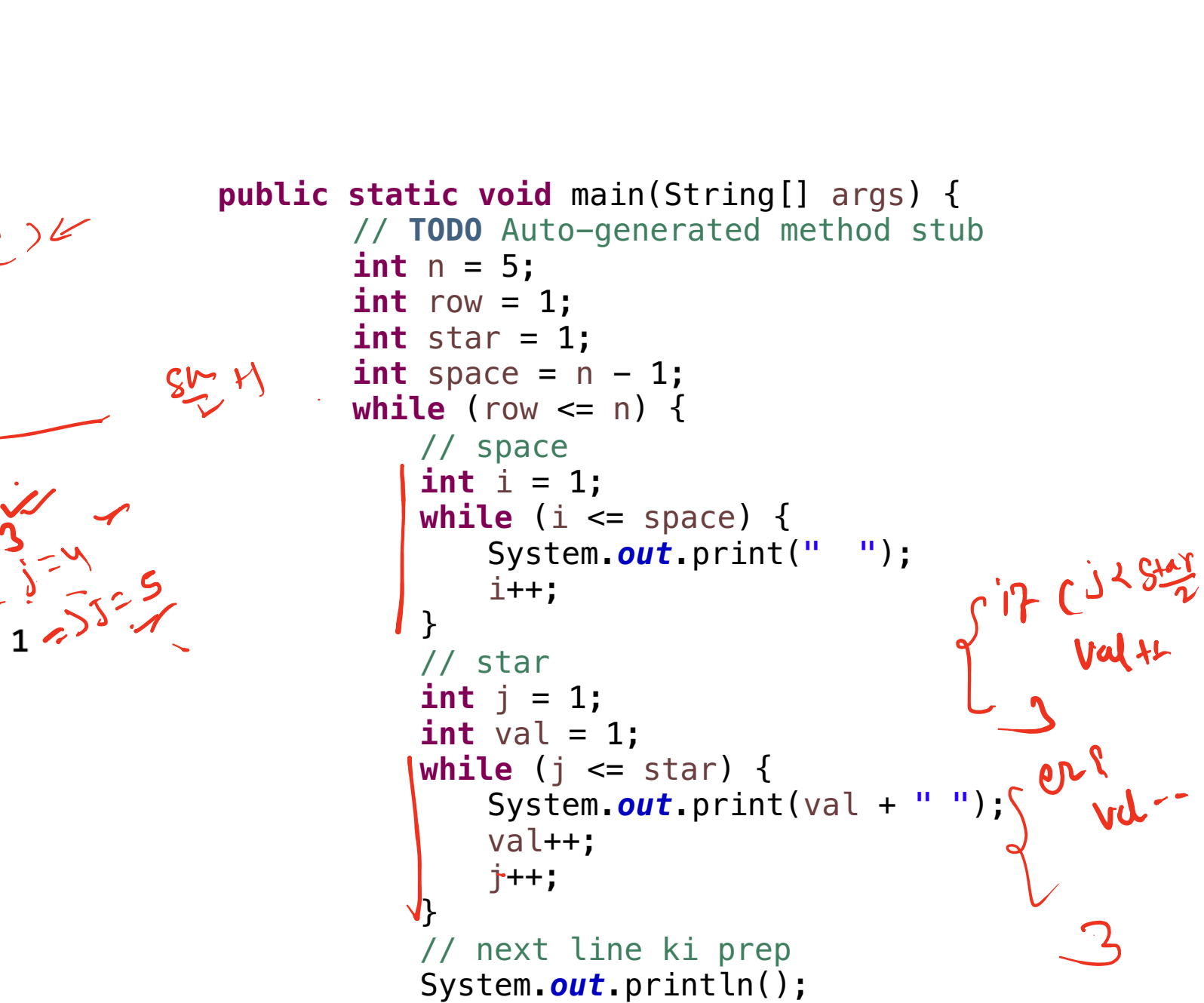
```
public static void main(String[] args) {
        // TODO Auto-generated method stub
        int n = 5;
        int row = 1;
        int star = 1;
        int space = n - 1;
        int val=1;
        while (row <= n) {
            // space
            int i = 1;
            while (i <= space) {
                System.out.print("  ");
                i++;
            }
            // star
            int j = 1;
            while (j <= star) {
                System.out.print(val+" ");
                val++;
                j++;
            }
            // next line ki prep
            System.out.println();
            row++;
            star += 2;
            space--;
        }

    }
```

Row=~~1~~ 2 3 4 5 Star=~~1~~ ~~3~~ 5 9 Space=4 ~~3~~ 2 1 0 val=~~1~~ 2 3 4 5 16 17



Ques 27:
n = 5

```
            1
         1  2  1
      1  2  3  2  1
   1  2  3  4  3  2  1
1  2  3  4  5  4  3  2  1
```

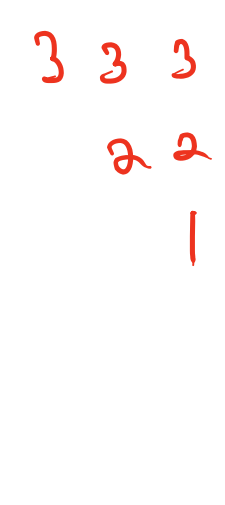


```
public static void main(String[] args) {
        // TODO Auto-generated method stub
        int n = 5;
        int row = 1;
        int star = 1;
        int space = n - 1;
        while (row <= n) {
            // space
            int i = 1;
            while (i <= space) {
                System.out.print("  ");
                i++;
            }
            // star
            int j = 1;
            int val = 1;
            while (j <= star) {
                System.out.print(val + " ");
                val++;
                j++;
            }
            // next line ki prep
            System.out.println();
            row++;
            star += 2;
            space--;
        }
    }
```

if (j < star/2 +1) {
val++
}
else {
val--
}

n=5

Row=1 space=n-1, star=1
while (Row<=2×n-1) {
// space
// star
if (Row < n) {
}
}

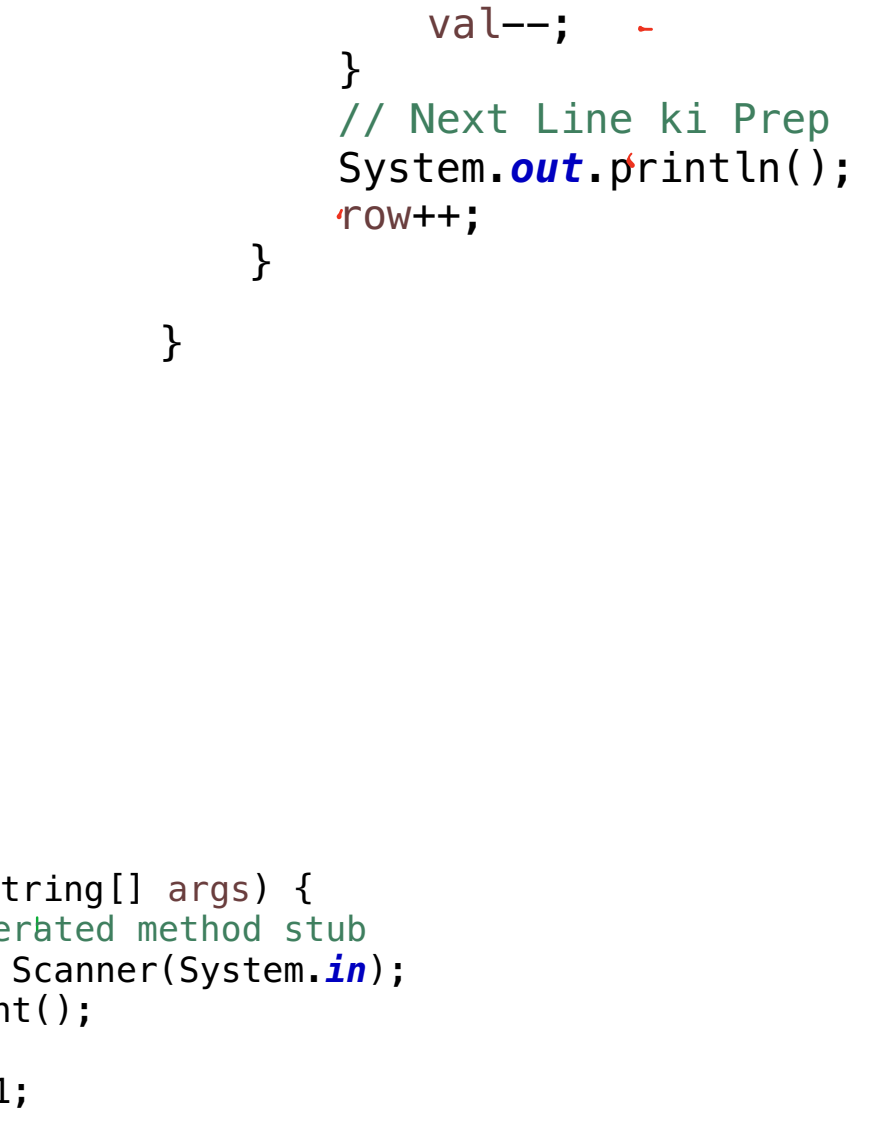


Take N (number of rows), print the following pattern (for N = 3).

```
    1
  2 3 2
3 4 5 4 3
  2 3 2
    1
```

Input Format

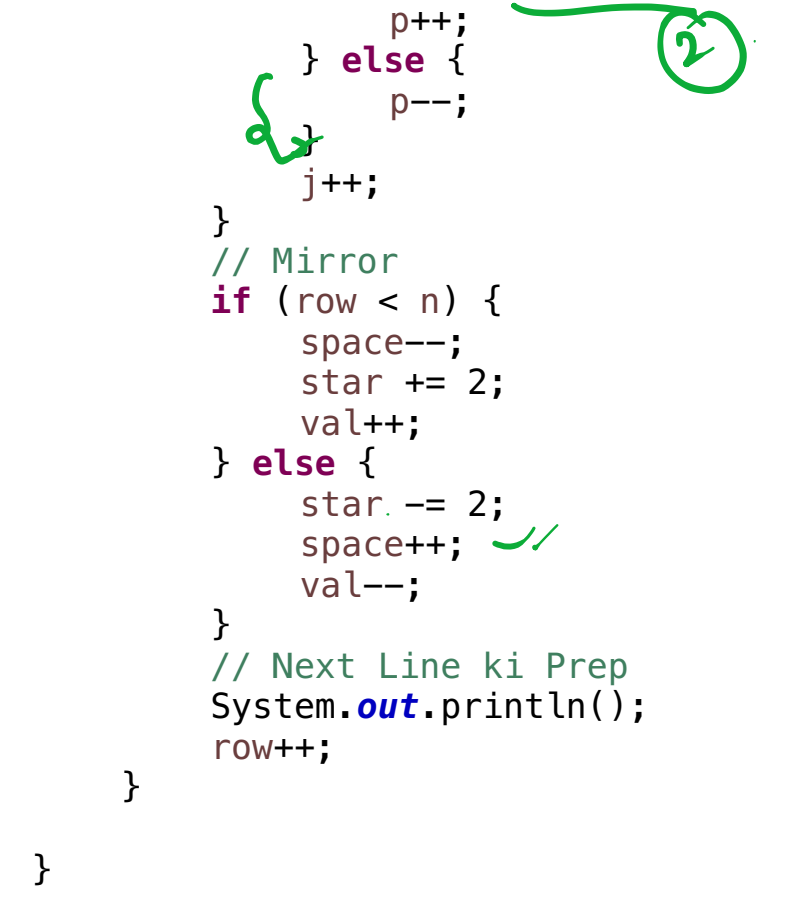
1
2 2 2
3 3 3 3 3
2 2 2
1



```
public static void main(String[] args) {
        // TODO Auto-generated method stub
        Scanner sc = new Scanner(System.in);
        int n = sc.nextInt();
        int row = 1;
        int space = n - 1;
        int star = 1;
        int val = 1;
        while (row <= 2 * n - 1) {
            // space
            int i = 1;
            while (i <= space) {
                System.out.print("\t");
                i++;
            }
            // star
            int j = 1;
            while (j <= star) {
                System.out.print(val + "\t");
                j++;
            }
            // Mirror
            if (row < n) {
                space--;
                star += 2;
                val++;
            } else {
                star -= 2;
                space++;
                val--;
            }
            // Next Line ki Prep
            System.out.println();
            row++;
        }

    }
```

N=3 val=~~1~~ 2 3 2 1 Row=~~1~~ 2 3 4 5 Space=~~2~~ ~~1~~ 0 1 2 Star=~~1~~ 3 5 3 1

2<3, 3<3



```
public static void main(String[] args) {
        // TODO Auto-generated method stub
        Scanner sc = new Scanner(System.in);
        int n = sc.nextInt();
        int row = 1;
        int space = n - 1;
        int star = 1;
        int val = 1;
        while (row <= 2 * n - 1) {
            // space
            int i = 1;
            while (i <= space) {
                System.out.print("\t");
                i++;
            }
            // star
            int j = 1;
            int p = val;
            while (j <= star) {
                System.out.print(p + "\t");
                if (j < star / 2 + 1) {
                    p++;
                } else {
                    p--;
                }
                j++;
            }
            // Mirror
            if (row < n) {
                space--;
                star += 2;
                val++;
            } else {
                star -= 2;
                space++;
                val--;
            }
            // Next Line ki Prep
            System.out.println();
            row++;
        }

    }
```

N=3 val=~~1~~ 2 3 2 1 Row=~~1~~ ~~2~~ 3 4 5 Space=~~2~~ 1 0 1 2 Star=~~1~~ 3 5 3 1